

Session: 3

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

- HTTP (HyperText Transfer Protocol) – Used for transferring web pages between browser and server.
- HTTPS (HyperText Transfer Protocol Secure) – Secure version of HTTP that encrypts data.
- FTP (File Transfer Protocol) – Used for transferring files between computers over a network.
- SMTP (Simple Mail Transfer Protocol) – Used for sending emails over the internet.

2. What is the difference between IPv4 and IPv6?

- IPv4 uses 32-bit addresses and supports fewer unique IP addresses.
- IPv6 uses 128-bit addresses and supports a much larger number of unique IP addresses.

3. Run nslookup  on your system and write down the IP address you get.

Command:

nslookup 

Example Output:

142.250.183.14

(Note: Your IP address may be different.)

4. What do these HTTP status codes mean?

- 200 – OK (Request successful)
- 301 – Moved Permanently
- 404 – Not Found
- 500 – Internal Server Error
- 403 – Forbidden

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

- Root – The top level of the DNS hierarchy (.)
- TLD – ".com"
- Domain – " "
- Subdomain – "www.google.com"

Example:

In "www.google.com", "www" is the subdomain, "google" is the domain name, and ".com" is the Top-Level Domain (TLD).